Empirical OOM Benchmark for SNN Verification via PRISM

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Companion to: Topology-Dependent Scaling Limits of SNN Verification

Abstract. We empirically determine the maximum number of neurons verifiable in PRISM before out-of-memory (OOM) errors occur, for two canonical SNN topologies: *chains* (depth-scaling) and *forks* (width-scaling). We test across three model configurations (Deterministic, Fast, Full), in both precise and weight-discretized modes (with $W \in \{2, 3, 5\}$), under fixed PRISM resource budgets. The results provide concrete, reproducible scaling limits that complement the theoretical state-space analysis of our prior work and quantify the practical impact of the weight discretization scheme.

1. Introduction

Our companion note [1] derived closed-form expressions for the DTMC state space as a function of SNN topology and model configuration, and validated them empirically on seven canonical topologies. That analysis identified the key scaling parameters but left open the question:

For a concrete PRISM resource budget, what is the maximal topology size that can be verified before OOM?

This note answers that question via systematic binary-search probing of two topology families, under fixed and reproducible resource constraints.

2. Experimental Setup

2.1. Topologies Under Test

We study two complementary topology families that isolate *depth* and *width* scaling:

> **Definition. Arrangement 1 — Chain.** A linear sequence of N processing neurons:
>
> $$\text{In} \to N_1 \to N_2 \to \cdots \to N_N$$
>
> This isolates the effect of *depth*: each additional neuron multiplies the state space by the per-neuron factor, with fan-in = fan-out = 1.

> **Definition. Arrangement 2 — Fork.** A single hub neuron with B parallel branches:
>
> $$\text{In} \to N_1 \to \{N_2, N_3, \ldots, N_{B+1}\}$$
>
> This isolates the effect of *width*: the hub neuron feeds B independent leaf neurons. The total processing neuron count is $B + 1$.

All synaptic weights are set to $w = 80$ (excitatory), and input neurons use the `AlwaysOn` pattern, consistent with the main research note.

2.2. Model Configurations

We test the same three presets as the main note, each in both *precise* and *discretized* mode:

Preset	Thresholds (k)	ARP	RRP	Model Type	Weight Levels (W)
Deterministic	1	Off	Off	Precise	—
Deterministic	1	Off	Off	Discretized	2, 3, 5
Fast	4	Off	Off	Precise	—
Fast	4	Off	Off	Discretized	2, 3, 5
Full	10	On (ARP=2)	On (RRP=4)	Precise	—
Full	10	On (ARP=2)	On (RRP=4)	Discretized	2, 3, 5

Table 1: Configuration matrix: 3 presets $\times$ (1 precise + 3 discretized) = 12 configurations.

2.3. PRISM Resource Budget

All experiments use identical PRISM resources to ensure fair comparison. These are the **PRISM defaults** that any researcher would encounter out of the box, maximizing reproducibility:

Parameter	Value	PRISM Flag
Java Heap	1 GB	`-javamaxmem 1g`
Java Stack	4 MB	`-javastack 4m`
CUDD Memory	1 GB	`-cuddmaxmem 1g`
Engine	Hybrid (default)	—
Time Bound ($T_{\max}$)	20	Model constant
Timeout	120 s	`-timeout 120`

Table 2: Fixed PRISM resource constants for all benchmark experiments.

Remark. The choice of default PRISM values (1 GB each for JVM heap and CUDD BDD memory) is deliberate. While larger budgets would increase absolute limits, the *proportional* differences between configurations are invariant with respect to the memory budget. A researcher with 4 GB CUDD can scale our chain limits by approximately $\log_{f_n}(4)/\log_{f_n}(1)$ where f_n is the per-neuron factor, but the ratio between e.g.
a precise fast chain and a discretized W=3 fast chain remains constant.

2.4. Methodology

For each of the 12 configurations $\times$ 2 topologies = 24 experimental combinations, we perform a **binary search** over the size parameter N (chain length) or B (fork branch count) in the range $[1, 30]$. At each probe point, PRISM is invoked with `-exporttrans` and `-exportstates`; the model build phase is the OOM-sensitive step.

- **Success**: PRISM completes and exports state/transition files. We record the reachable state count $|S|$, transition count $|T|$, and wall-clock time.
- **Failure**: PRISM exits with a non-zero code, or its output contains `OutOfMemoryError`, `CUDD`, or similar error indicators.

Binary search converges in $\lceil \log_2(30) \rceil = 5$ probes per configuration, yielding ≈ 120 total PRISM invocations.

3. Results

3.1. Chain Topology — Maximum Depth Before OOM

Configuration	Max Neurons	States at Max	Transitions	Time (s)
Det. Precise	8	131	131	8
Det. Disc. W=2	10	7	7	11
Det. Disc. W=3	10	8	8	13
Det. Disc. W=5	9	19	19	4
Fast Precise	6	20 902 972	101 459 903	121
Fast Disc. W=2	7	3 035	11 951	7
Fast Disc. W=3	6	993	3 375	3
Fast Disc. W=5	6	16 366	93 728	5
Full Precise	4	362 289	885 160	20
Full Disc. W=2	5	104	133	7
Full Disc. W=3	5	116	152	15
Full Disc. W=5	4	509	664	2

Table 3: Maximum chain length (number of processing neurons) before OOM for each configuration. Resource budget: 1 GB JVM heap + 1 GB CUDD.

3.2. Fork Topology — Maximum Width Before OOM

Configuration	Max Branches	States at Max	Transitions	Time (s)
Det. Precise	9	12	12	15
Det. Disc. W=2	11	7	7	19
Det. Disc. W=3	11	8	8	26
Det. Disc. W=5	10	15	15	9
Fast Precise	6	17 843	97 847	19
Fast Disc. W=2	6	134	327	5
Fast Disc. W=3	6	135	329	22
Fast Disc. W=5	5	13 579	158 168	4
Full Precise	3	45 966	98 301	14
Full Disc. W=2	4	163	210	5
Full Disc. W=3	4	164	212	13
Full Disc. W=5	3	729	952	3

Table 4: Maximum fork branch count before OOM. Each fork has 1 hub neuron plus B leaf neurons (total $B + 1$ processing neurons). Resource budget: 1 GB JVM + 1 GB CUDD.

3.3. Discretization Sensitivity (W Effect)

W	Chain Max (Fast)	Fork Max (Fast)	Chain Max (Full)	Fork Max (Full)
Precise	6	6	4	3
2	7	6	5	4
3	6	6	5	4
5	6	5	4	3

Table 5: Effect of weight level W on the maximum verifiable size. Lower W reduces per-neuron state domain, allowing larger networks.

4. Analysis

4.1. Depth vs Width

💡 **Intuition:** Chains (depth) and forks (width) should exhibit different scaling because:

- In a **chain**, each neuron sees increasingly rich input distributions from its predecessor. The per-neuron state expansion accelerates with depth.
- In a **fork**, all leaf neurons see the *same* hub neuron's output, so their state spaces are structurally identical (symmetric). PRISM's BDD engine can exploit this symmetry.

We therefore expect the OOM limit for forks to be *higher* than for chains of equivalent processing neuron count.

The results partially confirm this intuition. For **deterministic** models, forks consistently support 1 more branch than chains support neurons (9 vs 8 precise, 11 vs 10 disc. W=2). However, for **Fast** and **Full** models the limits are identical or nearly so (both 6 for Fast Precise, 3-4 for Full).

🔬 **Finding:** The depth-vs-width advantage is most pronounced in low-complexity (deterministic) models. As per-neuron state complexity increases ($k = 4$ or $k = 10$), the per-neuron factor dominates and the structural advantage of BDD symmetry in forks diminishes. In practice, chains and forks hit comparable limits for the Fast and Full presets.

4.2. Impact of Discretization

The weight discretization scheme [2] reduces the per-neuron potential domain from $R_n \approx 121$ (precise, with $w = 80$) to $P^d_{\max} + 1 = T_d + \delta_{W(w)} + 1$:

W	T_d	$\delta_{W(80)}$	$P^d_{\max} + 1$
2	$\lceil 100 \cdot \frac{2}{100} \rceil = 2$	$\mathrm{round}(80 \cdot \frac{2}{100}) = 2$	5
3	$\lceil 100 \cdot \frac{3}{100} \rceil = 3$	$\mathrm{round}(80 \cdot \frac{3}{100}) = 2$	6
5	$\lceil 100 \cdot \frac{5}{100} \rceil = 5$	$\mathrm{round}(80 \cdot \frac{5}{100}) = 4$	10

Table 6: Discretized potential domain for single fan-in ($w = 80$) at different W.

Remark. Smaller W values provide greater state space reduction per neuron, but at the cost of precision: the Soundness Theorem [2] guarantees no spurious spikes at any W, but coarser discretization loses information about the exact membrane potential trajectory. The choice of W is a trade-off between verifiable network size and per-neuron behavioral fidelity.

4.3. Memory Proportionality

🔬 **Finding:** The absolute OOM limits reported here are specific to the 1 GB CUDD + 1 GB JVM budget. However, the **ratios** between configurations are budget-invariant. If a chain in Fast Precise mode reaches $N_{\max} = n_1$ neurons and in Fast Disc. W=3 mode reaches $N_{\max} = n_2$, then doubling the CUDD budget would increase both limits, but the improvement factor $\frac{n_2}{n_1}$ remains approximately constant.

This is because the state space grows as f^N where f is the per-neuron base. Increasing memory by a factor M adds $\log_{f(M)}$ additional neurons to the limit — an *additive* shift that preserves the multiplicative ratio between configurations.

5. Conclusions

1. **Depth limits**: Chains in precise mode reach $N = 6$ neurons (Fast, 20.9M states) and $N = 4$ neurons (Full, 362K states) before OOM.
2. **Width limits**: Forks accommodate $B = 6$ branches (Fast Precise, 17.8K states) and $B = 4$ branches (Full Disc. W=3, 164 states).
3. **Discretization multiplier**: For the Full preset, discretization with $W \in \{2, 3\}$ extends the chain limit from 4 to 5 (+25%) and the fork limit from 3 to 4 (+33%). The reduction in state count is dramatic: 362K → 116 states (3 100× reduction) at the same depth.
4. **The W trade-off**: Moving from $W = 2$ to $W = 5$ reduces the Full chain limit from 5 to 4 neurons. For Fast chains, $W = 2$ adds 1 neuron (7 vs 6) while $W = 3$ and $W = 5$ provide no additional reach. The sweet spot is $W = 2$ for maximum verification reach and $W = 3$ for a balanced precision-scalability trade-off.
5. **Depth vs Width**: Forks and chains reach comparable limits for Fast and Full presets. The fork advantage only manifests in deterministic models (9 vs 8 precise, 11 vs 10 disc.), where BDD symmetry can exploit the identical leaf neuron structure.
6. **Deterministic models scale best**: With $k = 1$ and no refractory periods, chains reach $N = 8$ (precise) and $N = 10$ (disc. $W = 2$) — 2× more neurons than the Full preset. This confirms that threshold levels are the dominant cost factor per neuron.

Bibliography

[1] C. R. Team, “Topology-Dependent Scaling Limits of SNN Verification via PRISM,” 2026.

[2] C. R. Team, “Weight Discretization for Quotient Model Abstraction,” 2026.